

WORK EXPERIENCE

Southern University of Science and Technology Assistant Professor in Computer Science, Doctoral Supervisor	Shenzhen, China 2023–Current
Facebook Reality Labs Research Scientist at BioXR	Burlingame, United States Aug 2022 –Nov 2022

EDUCATION

Nanyang Technological University Ph.D. in Computer Science	Singapore 2019–2023
University of Toronto M.Eng. in Electrical and Computer Engineering	Toronto, Canada 2017–2019
Harbin Institute of Technology B.Eng. in Telecommunication Engineering	Harbin, China 2013–2017

GRANTS

Youth Fund of Natural Science Foundation of China	PI
General Project of Guangdong Basic and Applied Basic Research Foundation	PI
General Project of Shenzhen Science and Technology Innovation Program	PI
Enterprise-University Joint Algorithm Pre-research Program	PI
Huawei Outstanding Young Academic Talent Funding Project	PI

SELECTED PUBLICATIONS

(Author with “_” is student under my supervision, [†] denotes corresponding author, * denotes co-first author.)

1. Guoquan Wu, **Tianyue Zheng**[†], “SRay: Super-resolution Wireless Indoor Localization via Inverse Differentiable Ray-tracing”, *IEEE Transactions on Mobile Computing*, 2026, pp. 1–13. (CCF-A)
2. Hengwei Chen, Menglan Hu, and **Tianyue Zheng**[†], “Weaponizing Reflectivity for Pointcloud Deception with Forged Invisible Geometries”, in *Proc. of the 47th IEEE S&P*, 2026, pp. 1–18. (CCF-A)
3. Xueqiang Han, **Tianyue Zheng**[†], et al., “RayLoc: Wireless Indoor Localization via Fully Differentiable Ray-tracing”, in *Proc. of the 25th ACM UbiComp*, 2026, pp. 1–24. (CCF-A)
4. Xueqiang Han, **Tianyue Zheng**[†], and Jun Luo, “Stereo-Fi: Free-form 3D Reconstruction via Generatively Co-Trained Inverse RF Rendering”, in *Proc. of the 24th ACM SenSys*, 2026, pp. 1–24. (CCF-B)

5. Xueqiang Han, Jinyang Huang, Meng Li, Chao Cai, **Tianyue Zheng**[†], “Loki: Physical-World Adversarial Attacks on Wireless Indoor Localization via Differentiable Object Placement”, *IEEE Transactions on Information Forensics and Security*, 2025, pp. 13386 - 13400. (CCF-A)
6. Jinyi Cui and **Tianyue Zheng**[†], “E-M²: Efficient Multimodal Sensing via Adaptive Sensor-Computation Activation”, *IEEE Transactions on Mobile Computing*, 2025, pp. 1–12. (CCF-A)
7. Yuxuan Weng, **Tianyue Zheng**[†], et al., “Rising from Pieces: Effective Inference at the Edge via Robust Split ML”, *IEEE Transactions on Mobile Computing*, 2025, pp. 1–12. (CCF-A)
8. Yuxuan Weng, **Tianyue Zheng**[†], et al., “FM-Fi 2.0: Foundation Model for Cross-Modal Multi-Person Human Activity Recognition”, *IEEE Transactions on Mobile Computing*, 2025, pp. 1–12. (CCF-A)
9. Yuxuan Weng, Guoquan Wu, **Tianyue Zheng**[†], Yanbing Yang, and Jun Luo, “Large Model for Small Data: Foundation Model for Cross-Modal RF Human Activity Recognition”, in *Proc. of the 22nd ACM SenSys*, 2024, pp. 1–14. (CCF-B)
10. **Tianyue Zheng**, Jingzhi Hu, Rui Tan, Yinqian Zhang, Ying He, Jun Luo, “ π -Jack: Physical-World Adversarial Attack on Monocular Depth Estimation with Perspective Hijacking”, in *Proc. of the 33rd USENIX Security*, 2024, pp. 1–10. (CCF-A)
11. Jingzhi Hu*, **Tianyue Zheng***, Zhe Chen, Hongbo Wang, and Jun Luo, “MUSE-Fi: Contactless Multi-person SENSing Exploiting Near-field Wi-Fi Channel Variation”, in *Proc. of the 29th ACM MobiCom*, 2023, pp. 1–14. (CCF-A)
12. Shujie Zhang*, **Tianyue Zheng***, Zhe Chen, Jingzhi Hu, Abdelwahed Khamis, Jiajun Liu, and Jun Luo, “OCHID-Fi: Occlusion-Robust Hand Pose Estimation in 3D via RF-Vision”, in *Proc. of the IEEE/CVF ICCV*, 2023, pp. 1–10. (CCF-A)
13. **Tianyue Zheng**, Ang Li, Zhe Chen, Hongbo Wang, and Jun Luo, “AutoFed: Heterogeneity-Aware Federated Multimodal Learning for Robust Autonomous Driving”, in *Proc. of the 29th ACM MobiCom*, 2023, pp. 1–14. (CCF-A)
14. Shujie Zhang*, **Tianyue Zheng***, Hongbo Wang, Zhe Chen, and Jun Luo, “Quantifying the Physical Separability of RF-based Multi-Person Respiration Monitoring via SINR”, in *Proc. of the 20th ACM SenSys*, 2022, pp. 1–14. (CCF-B)
15. Ziwei Liu*, **Tianyue Zheng***, Chao Hu, Yanbing Yang, Yimiao Sun, Yi Zhang, Zhe Chen, Liangyin Chen, and Jun Luo, “CORE-Lens: Simultaneous Communication and Object REcognition with Disentangled-GAN Cameras”, in *Proc. of the 28th ACM MobiCom*, 2022, pp. 172–185. (CCF-A)
16. **Tianyue Zheng**, Zhe Chen, Shujie Zhang, and Jun Luo, “Catch Your Breath: Simultaneous RF Tracking and Respiration Monitoring with Radar Pairs”, *IEEE Transactions on Mobile Computing*, 2022, pp. 1–14. <https://ieeexplore.ieee.org/document/9852706/> (CCF-A)
17. **Tianyue Zheng**, Chao Cai, Zhe Chen, and Jun Luo, “Sound of Motion: Real-time Wrist Tracking with A Smart Watch-Phone Pair”, in *Proc. of the 41st IEEE INFOCOM*, 2021, pp. 1–10. (CCF-A)
18. **Tianyue Zheng**, Zhe Chen, Shujie Zhang, Chao Cai, and Jun Luo, “MoRe-Fi: Motion-robust and Fine-grained Respiration Monitoring via Deep-Learning UWB Radar”, in *Proc. of the 19th ACM SenSys*, 2021, pp. 1–14, <https://dl.acm.org/doi/abs/10.1145/3485730.3485932> (CCF-B)
19. **Tianyue Zheng**, Zhe Chen, Jun Luo, Lin Ke, Chaoyang Zhao, and Yaowen Yang, “SiWa: See into Walls via Deep UWB Radar”, in *Proc. of the 27th ACM MobiCom*, 2021, pp. 323–336. (CCF-A)

20. Zhe Chen*, **Tianyue Zheng***, and Jun Luo, “Octopus: A Practical and Versatile Wideband MIMO Sensing Platform”, in *Proc. of the 27th ACM MobiCom*, 2021, pp. 601–614. (CCF-A)
21. Zhe Chen*, **Tianyue Zheng***, Chao Cai, and Jun Luo, “MoVi-Fi: Motion-robust Vital Signs Waveform Recovery via Deep Interpreted RF Sensing”, in *Proc. of the 27th ACM MobiCom*, 2021, pp. 392–405, <https://dl.acm.org/doi/pdf/10.1145/3447993.3483251>. (CCF-A)
22. **Tianyue Zheng**, Zhe Chen, Shuya Ding, and Jun Luo, “Enhancing RF Sensing with Deep Learning: A Layered Approach”, *IEEE Communications Magazine*, vol. 59, no. 2, pp. 70–76, 2021, (SCI-Q1)
23. **Tianyue Zheng**, Zhe Chen, Chao Cai, Jun Luo, and Xu Zhang, “V²iFi: in-Vehicle Vital Sign Monitoring via Compact RF Sensing”, in *Proc. of the 20th ACM UbiComp*, 2020,70:1–27. (CCF-A)

TALK

1. “High-Performance Wireless Sensing driven by Physical AI and Foundation Models”, Huawei Wireless Professor Forum.
2. “Algorithmic Sensing for Energy-Efficient Green Edge Computing”, NRF FRC Green Computing Study Green Edge Workshop.

PROFESSIONAL SERVICES

- Executive Member
 - CCF Pervasive Computing Committee 2024 - Now
- TPC Member
 - IEEE INFOCOM, ACM MobiSys, IEEE ICPADS, IEEE MSN, IEEE UIC, IEEE/ACM CHASE, AIoTSys 2025
- Reviewer
 - IEEE TPAMI, ACM MobiCom, ACM UbiComp, AAAI, IEEE T-IFS, ICASSP, ACM ToSN, ACM ToIT, ...

TEACHING

- **Course Instructor** at Southern University of Science and Technology 2025 –Now
Advanced Computer Networks (CS5005)
- **Course Instructor** at Southern University of Science and Technology 2024 –Now
Computer Networks (CS305)
- **Course Instructor** at Southern University of Science and Technology 2023 –2024
Data Mining (CS306)
- **Teaching Assistant** at Nanyang Technological University 2019–2021
Computer Networks (CE3005/CZ3006) and Introduction to Computational Thinking (CE/CZ1003)
- **Teaching Assistant** at University of Toronto Spring 2019
Communication Systems (ECE316)

AWARDS

- World’s Top 2% Scientists 2025
- Shenzhen Peacock Talent Program (Category C) 2023
- Distinguished Reviewer, ACM Transactions on Sensor Networks (TOSN) 2026